

CCNA — Prompted Command Sequences

(Search by Topic)

Find a topic (e.g., Trunk, VLAN, OSPF). Under each, commands are listed in order with the exact prompt you should see. Replace values in < >.

VLAN — Create

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan <ID>
Switch(config-vlan)# name <NAME>
Switch(config-vlan)# exit
Switch(config)# end
```

Access Port — Assign to VLAN

```
Switch> enable
Switch# configure terminal
Switch(config)# interface range fa0/1 - 24 , gi0/1 - 2
Switch(config-if-range)# description <TEXT>
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan <ID>
Switch(config-if-range)# end
```

Trunk Ports (802.1Q)

```
Switch> enable
Switch# configure terminal
Switch(config)# interface range <PORTS>
Switch(config-if-range)# switchport trunk encapsulation dot1q
Switch(config-if-range)# switchport mode trunk
Switch(config-if-range)# switchport trunk allowed vlan <LIST>
Switch(config-if-range)# switchport trunk native vlan <ID>
Switch(config-if-range)# no shutdown
Switch(config-if-range)# end
```

EtherChannel — LACP Trunk

```
Switch> enable
Switch# configure terminal
Switch(config)# interface range <PORTS>
Switch(config-if-range)# shutdown
Switch(config-if-range)# channel-group 1 mode active
Switch(config-if-range)# exit
Switch(config)# interface port-channel 1
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan <LIST>
Switch(config-if)# no shutdown
Switch(config-if)# end
```

STP — Rapid-PVST + PortFast + BPDU Guard

```
Switch> enable
Switch# configure terminal
Switch(config)# spanning-tree mode rapid-pvst
Switch(config)# spanning-tree portfast default
Switch(config)# spanning-tree portfast bpduguard default
Switch(config)# end
```

Management SVI + Default Gateway

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan <MGMT-ID>
Switch(config)# interface vlan <MGMT-ID>
Switch(config-if)# ip address <IP> <MASK>
Switch(config-if)# no shutdown
Switch(config-if)# exit
Switch(config)# ip default-gateway <GATEWAY-IP>
Switch(config)# end
```

Interface IPv4 Address

```
Router> enable
Router# configure terminal
Router(config)# interface <TYPE/N0>
Router(config-if)# description <TEXT>
Router(config-if)# ip address <IPv4> <MASK>
Router(config-if)# no shutdown
Router(config-if)# end
```

Interface IPv6 Address (with global enable)

```
Router> enable
Router# configure terminal
Router(config)# ipv6 unicast-routing
Router(config)# interface <TYPE/N0>
Router(config-if)# ipv6 address <PREFIX>/<LEN>
Router(config-if)# no shutdown
Router(config-if)# end
```

DHCPv4 Server

```
Router> enable
Router# configure terminal
Router(config)# service dhcp
Router(config)# ip dhcp excluded-address <START-IP> <END-IP>
Router(config)# ip dhcp pool <NAME>
Router(dhcp-config)# network <NETWORK> <MASK>
Router(dhcp-config)# default-router <GW-IP>
Router(dhcp-config)# dns-server <DNS-IP>
Router(dhcp-config)# domain-name <DOMAIN>
Router(dhcp-config)# lease <DAYS>
Router(dhcp-config)# end
```

DHCP Relay (Helper) on Gateway Interface

```
Router> enable
Router# configure terminal
Router(config)# interface <LAN-UPLINK>
```

```
Router(config-if)# ip helper-address <DHCP-SERVER-IP>
Router(config-if)# end
```

NAT — PAT Overload (inside -> outside)

```
Router> enable
Router# configure terminal
Router(config)# interface <INSIDE-INT>
Router(config-if)# ip nat inside
Router(config-if)# exit
Router(config)# interface <OUTSIDE-INT>
Router(config-if)# ip nat outside
Router(config-if)# exit
Router(config)# ip access-list standard NAT-LAN
Router(config-std-nacl)# remark Permit inside LAN
Router(config-std-nacl)# permit <LAN-NET> <WILDCARD>
Router(config-std-nacl)# exit
Router(config)# ip nat inside source list NAT-LAN interface <OUTSIDE-INT> overload
Router(config)# end
```

Secure Management — Enable SSH, Users, VTY Lockdown

```
Router> enable
Router# configure terminal
Router(config)# security passwords min-length 10
Router(config)# enable secret <SECRET>
Router(config)# service password-encryption
Router(config)# banner motd $Unauthorized access strictly prohibited!$
Router(config)# hostname <NAME>
Router(config)# ip domain-name <DOMAIN>
Router(config)# username <USER> privilege 15 secret <PW>
Router(config)# line con 0
Router(config-line)# password <PW>
Router(config-line)# exec-timeout 10 0
Router(config-line)# login
Router(config-line)# logging synchronous
Router(config-line)# exit
Router(config)# line vty 0 4
Router(config-line)# transport input ssh
Router(config-line)# login local
Router(config-line)# exec-timeout 10 0
Router(config-line)# exit
Router(config)# crypto key generate rsa general-keys modulus 2048
Router(config)# ip ssh version 2
Router(config)# end
```

ACL — Standard (apply inbound on interface)

```
Router> enable
Router# configure terminal
Router(config)# ip access-list standard <NAME>
Router(config-std-nacl)# deny <NET> <WILDCARD> log
Router(config-std-nacl)# permit any
Router(config-std-nacl)# exit
Router(config)# interface <TYPE/NO>
Router(config-if)# ip access-group <NAME> in
Router(config-if)# end
```

ACL — Extended (template)

```
Router> enable
Router# configure terminal
Router(config)# ip access-list extended <NAME>
Router(config-ext-nacl)# deny <PROTO> <SRC-NET> <WILDCARD> <DST-NET> <WILDCARD> eq
Router(config-ext-nacl)# permit ip any any
Router(config-ext-nacl)# exit
Router(config)# interface <TYPE/NO>
Router(config-if)# ip access-group <NAME> in
Router(config-if)# end
```

Inter-VLAN — Router-on-a-Stick

```
Router> enable
Router# configure terminal
Router(config)# interface <ROUTER-LAN-INT>
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface <ROUTER-LAN-INT>.<VLAN-ID>
Router(config-subif)# encapsulation dot1q <VLAN-ID> [native]
Router(config-subif)# ip address <IP> <MASK>
Router(config-subif)# end
```

L3 Switch — SVIs Routing

```
Switch> enable
Switch# configure terminal
Switch(config)# ip routing
Switch(config)# interface vlan <X>
Switch(config-if)# ip address <IP> <MASK>
Switch(config-if)# exit
Switch(config)# interface vlan <Y>
Switch(config-if)# ip address <IP> <MASK>
Switch(config-if)# end
```

Static Default Route (IPv4)

```
Router> enable
Router# configure terminal
Router(config)# ip route 0.0.0.0 0.0.0.0 <NEXT-HOP-IP>
Router(config)# end
```

OSPFv2 — Minimal Working Config

```
Router> enable
Router# configure terminal
Router(config)# router ospf 1
Router(config-router)# router-id <A.B.C.D>
Router(config-router)# network <NET> <WILDCARD> area 0
Router(config-router)# end
```

Troubleshooting — Quick Show Commands

```
Router> enable
Router# show ip interface brief
Router# show interfaces <INT>
Switch# show vlan brief
```

```
Switch# show interfaces trunk
Router# show cdp neighbors detail
Router# show lldp neighbors
Router# show ip route
Router# show arp
Router# show logging
```